

Paras Rimal

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🐙 github.com/ParasRimal

Driven learner focused on AI, machine learning, and database systems. Passionate about understanding the mathematical principles and internal workings of ML algorithms, with a keen interest in applying data-driven solutions to industry projects.

Education

2024 – 2028 Maitidevi, Kathmandu	BSc (Honors) Computer Science with Artificial Intelligence Sunway College Kathmandu 🔗
2022 – 2024 Balkumari, Lalitpur	Higher Secondary Education (+2) in Science Kathmandu Model Secondary School 🔗

Skills

Technical Skills

- Programming Languages: Python, JavaScript, SQL
- AI & ML: ML Algorithms (Decision Trees, Regression, Classification), Model Evaluation & Drift Detection
- APIs & Web: FastAPI, Django, REST APIs, React

Data & Tools

- Databases & CS Foundations: MySQL, Relational Database Design, DSA
- MLOps & Pipeline Tools: Docker, Apache Airflow (DAGs), MLflow, Evidently AI, Grafana

Interests

- **AI/ML & Mathematics:** Understanding model foundations, algorithms, and how they work beneath abstractions.
- **Prompt Engineering & LLMs:** Exploring effective prompting, reasoning, and first-principles problem solving.
- **Database Systems & Data Engineering:** Design, indexing, query optimization, and efficient data management.
- **DSA & Optimization:** Time vs space complexity, performance, resource efficiency, and creative problem solving.

Projects

Heart Disease Prediction using ML (Random Forest and Logistic Regression) [🔗](#)

- Built and evaluated predictive models (Logistic Regression and Random Forest) to classify heart disease risk using clinical and lifestyle data.
- Performed data preprocessing, EDA, feature engineering, and developed a user-input prediction interface with performance comparison using Accuracy, F1-score, and ROC-AUC.

Smart Grid Peak Load Risk Prediction (Apache Airflow · MLflow · FastAPI · Evidently AI · Grafana · Docker · MariaDB ColumnStore · Redis) [🔗](#)

- Engineered a Dockerized production-grade MLOps pipeline with Apache Airflow orchestrating hourly training of 4 models (Random Forest, XGBoost, LightGBM, Gradient Boosting), registering the champion in MLflow and serving real-time predictions via FastAPI REST API with a custom web UI.
- Deployed Docker containers for MariaDB ColumnStore (data warehouse) and Redis (real-time cache), integrated Evidently AI for automated drift detection and retraining, and built Grafana dashboards — achieving 90%+ accuracy through composite target engineering.

Bus Reservation System (PHP, MySQL, JavaScript, Bootstrap) [🔗](#)

- Developed a comprehensive booking platform using PHP and MySQL, featuring real-time trip scheduling, a 2x2 visual seat selection map, and automated PNR-based e-ticket generation.
- Engineered a responsive multi-tenant architecture with HTML, CSS, Bootstrap, and JavaScript, ensuring strict data isolation for independent operators and an intuitive interface for passengers.

Certifications & Achievements

Birmingham City University	Bronze Award - BCU Graduate+ 🔗 <i>Recognized for achievement in professional development and employability skills.</i>
Sunway College Kathmandu	Git & React 🔗 Built a personal portfolio website using the React framework.
Programiz	Learn Python Basics 🔗 Developed small projects demonstrating loops, functions, and data handling.